Project Framework

Colab Link ➕

https://colab.research.google.com/drive/11xVTa87KZOLFaPxKEslcXY6WhT-RgUTf?usp=sharing

https://colab.research.google.com/drive/1PFDqafwTkrx3ILhW1NYYI_f_B73XhGyy?usp=sharing

Colab

Phase 1 — Problem Framing

- Label structure
- Imbalance check
- Data modality

Phase 2 — EDA & Audit

- Plot class distribution-visualize
- total samples per split (train/val/test)
- Check input shape consistency
- min, max, mean, std per channel/feature (value range check)
- label corresponding files
- temporal or patient-level leakage between train and val splits

Phase 3 — Data Cleaning

- Validate label encoding
- Check for corrupted files
- label noise checking

Phase 4 — Class Imbalance Strategy

Severity Guide

Ratio means how many more samples the biggest class has compared to the smallest.

Ratio	Severity	Approach
1:1 – 1:2	Balanced	Standard CE loss, nothing extra
1:2 – 1:10	Mild	Weighted CE as a quick baseline
1:10 – 1:100	Moderate	Focal Loss + WeightedRandomSampler
1:100+	Severe	Focal Loss (high gamma) + Sampler + minority augmentation

We will first use focal use.

Phase 5 — Preprocessing

`` Resize to match model's expected input — 224×224

`` Normalization for natural images.

Augmentation on training set only

Phase 6 — Dataset & DataLoader

Phase 7 — Model Architecture

Approach will be finalise later on

Phase 8 — Loss Function

Phase 9 — Optimizer & Scheduler

Phase 10 — Training Loop

Phase 11 — Evaluation

Phase 12 — Error Analysis